Adnan Harun Doğan

Personal Information

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Research Experience

Research Assistant, Image Processing and Pattern Recognition Laboratory (ImageLab) — Feb 2023–Present
Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *METU, Ankara, Turkey*
- Contributed to Bucketed Ranking Loss, training DETR/transformer detectors up to 6x faster (ECCV 2024)
- Designed optimal-transport (Sinkhorn-Knopp) and ranking losses for end-to-end DETR-style object detection

Research Intern, Sensing, Interaction, & Perception Laboratory (SIPLab) — Apr 2025–Sep 2025
Supervisor: Prof. Dr. Christian Holz — *ETH Zürich, Switzerland*
- Trained Kalman filters to denoise the innovation spectrum under spectral supervision for colored noise
- Built EEG ConvLSTM models for real-time VR cybersickness detection on spectra, +30% over baselines

Intern, Center for Artificial Intelligence and Robotics (ROMER) — Oct 2021–Feb 2022
Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku — *METU, Ankara, Turkey*
- Built an RC-car autonomous-driving stack (CarRL) and trained RL navigation agents in ROS2/Gym/PyTorch
- Approximated multi-agent localization solvers for a Hector-quadrotor drone swarm simulated in ROS/Gazebo

Research Intern, Cyber-Physical Systems Laboratory (AICPS) — Jun 2021–Sep 2021
Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque — *UC Irvine, California, USA*
- Built multi-modal deep models detecting cardiac and sleep abnormalities from ECG and physiological signals
- Fused time-series and spectral features into challenge-competitive ECG arrhythmia classifiers (CinC 2021)

Education

Master of Science in Computer Engineering — 2022–2025
Middle East Technical University (METU) — *Ankara, Turkey*
Advisor: Prof. Dr. Sinan Kalkan, **Co-advisor:** Assoc. Prof. Dr. Emre Akbas
Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation

Bachelor of Science in Computer Engineering — 2018–2022
Middle East Technical University (METU) — *Ankara, Turkey*
Senior Project: ReQuest: Automated Question-Answer Generation from Text

Selected Graduate Coursework

Optimization: Linear Optimization (IE 553), Numerical Optimization (IAM 566), Optimization Methods for ML (IAM 771)
Probability and Statistics: Probability Theory (IAM 541), Decision Theory and Bayesian Analysis (STAT 565)
Deep Learning: Deep Learning (CENG 501), Advanced Deep Learning (CENG 502), Deep Generative Models (CENG 796)

Research Interests

My research has focused on gradient-based learning through non-differentiable operations. I am interested in problems where the model or its loss contains a discrete decision (e.g., ordering detections by confidence, assigning predictions to targets, choosing which layers a token passes through, or deciding where to refine a mesh). Such decisions give backpropagation little or no gradient signal. I have designed structure-respecting surrogate gradients for the building blocks behind them, such as ranking, sorting, matching and routing, and studied how this choice shapes what the network learns. Continuous relaxations of these decisions also lead me to optimal transport and measure-theoretic tools.

Keywords: gradients through discrete decisions, differentiable optimization, decision-focused learning, optimal transport, neural state estimation, differentiable ranking and matching, conditional computation

Publications

Selected publications

5. Dogan, A. H., Demirel, B. U., & Holz, C. (2027). *Learned-noise Kalman filtering with a body prior for drift-calibrated full-body pose from raw sparse inertial signals* [Manuscript submitted for publication][†].
4. Dogan, A. H., Arasli, D., Akbas, E., Kalkan, S., & Oksuz, K. (2027). *Variational surrogate Jacobians: A framework for constrained gradient propagation* [Manuscript submitted for publication][†].
3. Dogan, A. H., Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. In ***2026 IEEE International Conference on Robotics and Automation (ICRA)***. IEEE. https://doi.org/10.48550/arXiv.2606.02251 (PDF, Code)
2. Demirel, B. U., Dogan, A. H., Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE Transactions on Visualization and Computer Graphics, 31***(5), 2525–2534. https://doi.org/10.1109/TVCG.2025.3549132 (arXiv, Code)
1. Yavuz, F., Cam, B. C., Dogan, A. H., Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. In ***Computer Vision – ECCV 2024*** (Lecture Notes in Computer Science, Vol. 15117, pp. 93–109). Springer. https://doi.org/10.1007/978-3-031-73202-7_6 (arXiv, PDF, Code)

Other publications

8. Dogan, A. H., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows* [Manuscript submitted for publication].
7. Dogan, A. H., Deniz, M., Alemdar, H., & Baran, O. U. (2026). *CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws.* arXiv. https://doi.org/10.48550/arXiv.2607.06587
6. Amin, M. A., Dogan, A. H., Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. In *Advanced Information Networking and Applications* (Lecture Notes on Data Engineering and Communications Technologies, Vol. 204, pp. 217–226). Springer. https://doi.org/10.1007/978-3-031-57942-4_22 (Data)
5. Karagoz, P., Cekinel, R. F., Dogan, A. H., Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. In M. Akkar Ercan & K. Aydiner (Eds.), *Valorising Underground Built Heritage in Cappadocia* (pp. 61–92). CNR Edizioni. ISBN 978-88-8080-605-9. (PDF)
4. Sever, Y., & Dogan, A. H. (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies, 4*(2), 383–388. https://doi.org/10.52953/fplr8631 (PDF, Code)
3. Sever, Y., Ekinci, G., Dogan, A. H., Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. In *2022 International Workshop on Secure and Reliable Microservices and Containers (SRMC)* (pp. 18–26). IEEE. https://doi.org/10.1109/SRMC57347.2022.00007 Best Paper Award
2. Demirel, B. U., Dogan, A. H., & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. In *2021 Computing in Cardiology (CinC)* (pp. 1–4). IEEE. https://doi.org/10.23919/cinc53138.2021.9662935 (PDF, Code)
1. Buyukbas, E. B., Dogan, A. H., Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. In *Big Data Analytics and Knowledge Discovery* (Lecture Notes in Computer Science, Vol. 12925, pp. 152–157). Springer. https://doi.org/10.1007/978-3-030-86534-4_14

Theses

1. Dogan, A. H. (2025). *Incorporating piecewise linear functions with constant regions in backpropagation* **[Master's thesis, Middle East Technical University]**. (Thesis)

[†] Venue withheld while under double-blind review.

Undergraduate Research Experience

Container Security (CONTSEC), METU Sep 2021–Jun 2022
Supervisor: Assoc. Prof. Dr. Pelin Angin *Ankara, Turkey*
Project: High Performance Intrusion Detection and Prevention for Software-Defined Container Networks in the Cloud (TUBITAK 3501)
Topic: Security of containerized microservices on Kubernetes

- Compared ML intrusion detectors on system-call and network-flow features of Kubernetes containers
- Built a public Kubernetes misuse dataset of network flows from 10 labelled attack scenarios (ITU JFET 2023)

Image Processing and Pattern Recognition Laboratory (ImageLab), METU Feb 2021–Jun 2022
Supervisor: Assist. Prof. Dr. Hande Alemdar, Assist. Prof. Dr. Ozgur Ugras Baran *Ankara, Turkey*
Topic: Conservation flux informed neural networks and adaptive mesh refinement for transonic airfoils

- Designed an HLLC Riemann solver momentum flux loss on airfoil flow pixels, cutting drag error by 15%
- Trained CNN, FNO, ViT and diffusion models with and without the momentum flux loss on transonic flows
- Refined transonic GNN airfoil meshes at Ducros-flagged shocks, cutting the area-based RMSE by 12%

Textual Event Graph HUB (TEGHub), METU Oct 2020–Jun 2021
Supervisor: Prof. Dr. Pinar Karagoz *Ankara, Turkey*
Topic: Textual analysis for irony detection and research impact prediction

- Adapted T5 to Turkish irony detection, its first use for the task, reaching 59.5% accuracy (DaWaK 2021)
- Fine-tuned BERT with Simple Transformers, reaching 69.0% accuracy on Turkish irony detection data
- Crawled scientific papers and their Twitter mentions as training data for research impact prediction models
- Applied LDA, K-Means and rule-based VADER sentiment to 1,742 reviews of Cappadocia's underground sites

School of Medicine, Koç University Feb 2020–Jun 2021
Supervisor: Assist. Prof. Dr. Ozge Karanfil *Istanbul, Turkey*
Topic: Healthcare policy simulation for evidence-based decision making

- Used simulation models of mechatronics applications in healthcare for evidence-based policy making
- Built the citation graph of 999 health systems research papers from Microsoft Academic Graph (MAG) data
- Developed GCN-based node embeddings on this citation graph of the health systems research literature

Teaching Experience

Teaching Assistant, Computer Engineering Department, METU Spring 2023–Present

8. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025, Fall 2026)
7. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
6. CENG 242 - Programming Language Concepts (Spring 2023, Spring 2025, Spring 2026)
5. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)
1. CENG 334 - Introduction to Operating Systems (Spring 2023)

Undergraduate Teaching Assistant (Volunteer), Computer Engineering Department, METU Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)
2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Honors and Scholarships

2022–2023	TUBITAK 2247-C STAR'23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2021–2022	TUBITAK 2247-C STAR'22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR'21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Computer Skills

Languages:	Python, C/C++, Rust
Tools & Libraries:	PyTorch, CVXPYLayers, Gurobi, OR-Tools, POT, MNE, ROS2
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)